

CHIH-HAO CHOU

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EDUCATION

B.S. in Mechanical Engineering , National Taiwan University (NTU)

Aug. 2021 - Expected Jun. 2026

- Overall GPA: 4.12/4.30 (3.93/4.0)
- Relevant Coursework: Linear Systems, Adaptive Control, Digital Control, Robotics, Robot Vision, Algorithm
- Certificates of Specialization Program (each requires completion of 4-5 specific courses):
Robotics, Artificial Intelligence, Design & Practice of Electric Vehicles, System Control & Mechatronics

RESEARCH EXPERIENCE

2-DOF Dual-Mode Pendulum-Driven Spherical Robot

Aug. 2023 - Sep. 2025

Bio-Inspired Robotics Lab, NTU

Advised by Prof. Pei-Chun Lin

- Developed a spherical robot capable of omnidirectional locomotion and self-rescue on soft terrain using bevel gear
- Established kinematic and dynamic models using Lagrangian mechanics to simulate motion profiles
- Designed modular frame and integrated hierarchical embedded system (Raspberry Pi/Arduino) with IMU feedback
- Validated performance using Vicon motion capture system, confirming an 87.1% reduction in tracking error and a 7.4x faster escape time on soft terrain

Data-Driven Kinematic Stability Assessment for Artificial Joints

Aug. 2024 - Sep. 2025

Smart & Advanced Manufacturing Lab, NTU

Advised by Prof. Dian-Ru Li

- Established a standardized benchmark for small joint stability by developing a reproducible dynamic testing platform that integrates force sensors with computer vision analysis
- Designed an analytical framework to quantify strain in critical regions: tracked motion via optical flow, segmented zones using PCA-reduced kinematic features and K-means clustering, and computed Green-Lagrange tensors
- Derived indices for local strain and reaction force attenuation, discovering inverse relationship with surgical defects

Street Sweeping Mobile Robot

Sep. 2023 - Jun. 2024

Intelligent Vehicle & Mechatronics Lab, NTU

Advised by Prof. Kang Li

- Implemented SLAM in campus environments using Fast-LIO for 3D map generation and NDT for re-localization
- Applied Open3D-ML for semantic segmentation of 3D LiDAR point clouds using KPConv and RandLA-Net
- Fine-tuned a PointPillars model using custom-labeled 3D LiDAR data, boosting real-world performance by 10% while maintaining real-time inference

PUBLICATIONS

C=CONFERENCE, J=JOURNAL

- [J.1] P. Y. Chang*, **C. H. Chou***, Y. P. Chang*, and P. C. Lin, "**Development of a Dual-Mode Spherical Robot Using a Differential Drive**," in *IEEE Robotics and Automation Letters*, Under Review, 2025 (*Co-first author)
- [J.2] C. Y. Hong*, **C. H. Chou***, K. L. Wu, Y. J. Wang, and D. R. Li, "**Development of a Smart Cadaveric Limb Motion Simulator for Physiologically Relevant In-Vitro Biomechanical Evaluation of Small Joints**," in *IEEE Transactions on Biomedical Engineering*, Under Review, 2025 (*Co-first author)
- [C.1] P. Y. Chang*, **C. H. Chou***, T. J. Wang*, Y. P. Chang*, and P. C. Lin, "**A Two-Degree-of-Freedom Pendulum-Driven Spherical Robot Platform**," *2025 International Conference on Advanced Robotics and Intelligent Systems (ARIS)*, Taipei, Taiwan, 2025 (*Co-first author)
- [C.2] K. L. Wu*, C. Y. Hong*, **C. H. Chou***, Y. J. Wang*, and D. R. Li, "**Smart Joint Motion Simulation Platform with Integrated Force Sensing for Clinical Assessment of Artificial Finger Joints**," *CSME 2025 Annual Meeting and the 42nd National Academic Conference*, Taipei, Taiwan, 2025 (*Co-first author)

WORK EXPERIENCE

Technology Development Intern

Taipei, Taiwan

URS Robot Inc.

Jun. 2024 - Aug. 2024

- Built a ROS 2 turf monitoring system combining AI-based segmentation with GNSS to generate high-res heatmaps
- Developed an auto-labeling pipeline using ExG index and K-means clustering, producing high-quality training data
- Designed dual U-Net models to segment green grass, withered grass, and ground, improving accuracy under varying lighting conditions

Software Engineer Intern

Hsinchu, Taiwan

Industrial Technology Research Institute (ITRI)

Jul. 2023 - Feb. 2024

- Developed a program in ROS 2 using 64px ToF sensors and DBSCAN algorithm to differentiate planes
- Merged point clouds from 2 2D LiDARs, enabling detection of nearby obstacles
- Prevented AMR from falling at harbor edges and warned against close obstacles for safe operation

Calculus Teaching Assistant

Taipei, Taiwan

Department of Mathematics, NTU

Feb. 2023 - Jun. 2023

- Delivered lectures, graded coursework, and supported students in an English-Mediated (EMI) Calculus course

PROJECTS

Aero Rider (Capstone Project) Practice of Mechanical Engineering, NTU	Taipei, Taiwan Feb. 2024 - Jun. 2024
• Led a 5-person team to design, build, and test an autonomous wind-powered vehicle to navigate defined wind fields • Developed a PD controller that utilizes IMU data to modulate dual sail surface area for vehicle orientation • Conducted structural FEA and aerodynamic analysis for sail sizing, validated results through iterative testing	
Stair-Climbing Robot Introduction to Robotics, NTU	Taipei, Taiwan Oct. 2024 - Dec. 2024
• Designed a dual-chassis mobile robot connected by an RRR manipulator for stair climbing and human-following • Integrated a three-depth-sensor array to detect relative human position, enabling real-time following behavior • Generated smooth manipulator trajectories via cubic polynomial interpolation between defined kinematic key poses	
Robotic BackFlip Cat 2023 MakeNTU	Taipei, Taiwan May 2023
• Engineered a spring-loaded leg mechanism utilizing a motor-winch system to compress springs for energy storage • Integrated carbon fiber tubes with 3D-printed components to reduce limb inertia and maximize launch acceleration	

HONORS AND AWARDS

Presidential Award , NTU	Academic Year 2023–2024
• Earned full-tuition scholarship for 2 semesters for being in top 2% of students in department	
Dean’s List Award (×4) , NTU	Spring 2022, Fall 2022, Fall 2024, Spring 2025
• Recognized 4 times for achieving GPA in top 5% of class and received 24,000 NTD in total	
Brilliant TA Award , NTU	Spring 2023
• Earned recognition and prize of 5,000 NTD based on student evaluations	
1st Place , 2024 Mechaheroes Intercollegiate Engineering Project Competition	Dec. 2024
• Won out of 27 teams with “2-DOF Dual-Mode Pendulum-Driven Spherical Robot,” earning 30,000 NTD	
Micron LAUNCH Award , Tokyo Tech Engineering Sustainability Challenge	Dec. 2023
• Earned team award, including 300,000 JPY in development funds from Micron Foundation	
Best Popularity Award , 2023 MakeNTU: "Robotic BackFlip Cat"	May 2023
• Voted by participants as most creative and engaging project among 40 competing teams	

RELEVANT SKILLS

Programming Languages: C++, Python, MATLAB
Robotics & AI Technologies: ROS 2, OpenCV, Pytorch, Simulink
CAD & Design Tools: AutoCAD, Autodesk Inventor, SolidWorks

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Vice-President of NTUME Student Association National Taiwan University	Taipei, Taiwan Jul. 2023 - Jun. 2024
• Established dedicated study spaces within the department, improving the learning environment for undergraduates • Directed large-scale student events to foster community engagement and professional networking within NTUME	
Participant, Tokyo Tech Engineering Sustainability Challenge Institute of Science Tokyo formerly Tokyo Tech	Online/Tokyo, Japan Aug. 2023 - Dec. 2023
• Collaborated with a cross-national team (NTU & Tokyo Tech) to develop a business proposal for a drone-based robotic system addressing labor shortages in aging communities • Proposed a solution to optimize garbage collection efficiency and enhance accessibility for elderly residents	
Participant, Google Hardware Product Sprint Google LLC	Taipei, Taiwan Jun. 2023 - Aug. 2023
• Collaborated with a cross-functional team to develop LOCUS, a lockbox for timed focus with distinct study modes • Developed OpenCV face detection to trigger water-mist deterrent and auto-log compliance photos on early unlocks • Created immersive study modes with distinct lighting, music, wallpapers, and scents, switchable via hand gestures	
Participant, International Companions for Learning Ministry of Education Republic of China (Taiwan)	Online/Taoyuan, Taiwan Sep. 2021 - Jan. 2022
• Co-hosted weekly sessions with international partners to foster cultural exchange for rural Taiwanese students • Facilitated in-person cross-cultural interactions and language learning during semester field trips to remote areas	